

SARASWAT ACADEMY
TEST- COORDINATE GEOMETRY
(CLASS-X)

Section A – Multiple Choice Questions (1 mark each × 4 = 4 marks)

- Q1: The point which divides the line segment of points P (-1, 7) and (4, -3) in the ratio of 2:3 is:
(a) (-1, 3) (b) (-1, -3) (c) (1, -3) (d) (1, 3)
- Q2: The distance of the point (α, β) from the origin is:
(a) $\alpha + \beta$ (b) $\alpha^2 + \beta^2$ (c) $|\alpha| + |\beta|$ (d) $\sqrt{\alpha^2 + \beta^2}$
- Q3: The points A(9,0), B(9, -6), C(-9,0) and D(-9,6) are the vertices of a
(A) Square (B) Rectangle (C) Parallelogram (D) Trapezium
- Q4: The coordinates of a point P, where PQ is the diameter of a circle whose centre is $(2, -3)$ and Q is $(1, 4)$ is:
(a) $(3, -10)$ (b) $(2, -10)$ (c) $(-3, 10)$ (d) $(-2, 10)$

Section B – Short Answer Questions (2 marks each × 3 = 6 marks)

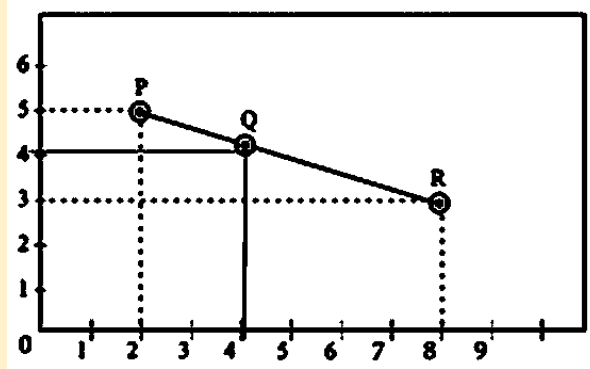
- Q5: Let P and Q be the points of trisection of the line segment joining the points A $(2, -2)$ and B $(-7, 4)$ such that P is nearer to A. Find the coordinates of P and Q.
- Q6: Write the coordinates of a point on the x-axis which is equidistant from points A $(-2, 0)$ and B $(6, 0)$.
- Q7: The x-coordinate of a point P is twice its y-coordinate. If P is equidistant from Q $(2, -5)$ and R $(-3, 6)$, find the coordinates of P.

Section C – Long Answer Questions (3 marks each × 2 = 6 marks)

- Q8: If A $(-2, 1)$, B $(a, 0)$, C $(4, b)$ and D $(1, 2)$ are the vertices of a parallelogram ABCD, find the values of a and b. Hence find the lengths of its sides.
- Q9: Prove that the points $(3, 0)$, $(6, 4)$ and $(-1, 3)$ are the vertices of a right-angled isosceles triangle.

Section D – Case Study Based Question (4 marks)

A group of class X students goes to picnic during winter holidays. The position of three friends Aman, Kirti and Chahat are shown by the points P, Q and R



- (i) Find the distance between P and R.
- (ii) Is Q, the midpoint of PR? Justify by finding midpoint of PR.
- (iii) (A) Find the point on x-axis which is equidistant from P and Q.

OR

(B) Let S be a point which divides the line joining PQ in ratio 2:3. Find the coordinates of S.